

Chapter 1

Atomic Structure

Learning Objectives

- Understand basic chemistry concepts: pure substance, mixture, physical/chemical property, intensive/extensive property
- Know the brief development of atomic model
- Describe the Schrodinger Wave equation; orbitals, spin number; Hund's Rule
- Explain isotopes and atomic mass, atomic number
- Recognise the s, p, d, f orbitals of an atom
- Recognise the shape of the s, p and d orbitals
- Apply Pauli's Exclusion principle, Aufbau's Principle and Hund's rule for determining the electronic configuration of an atom

Part 1

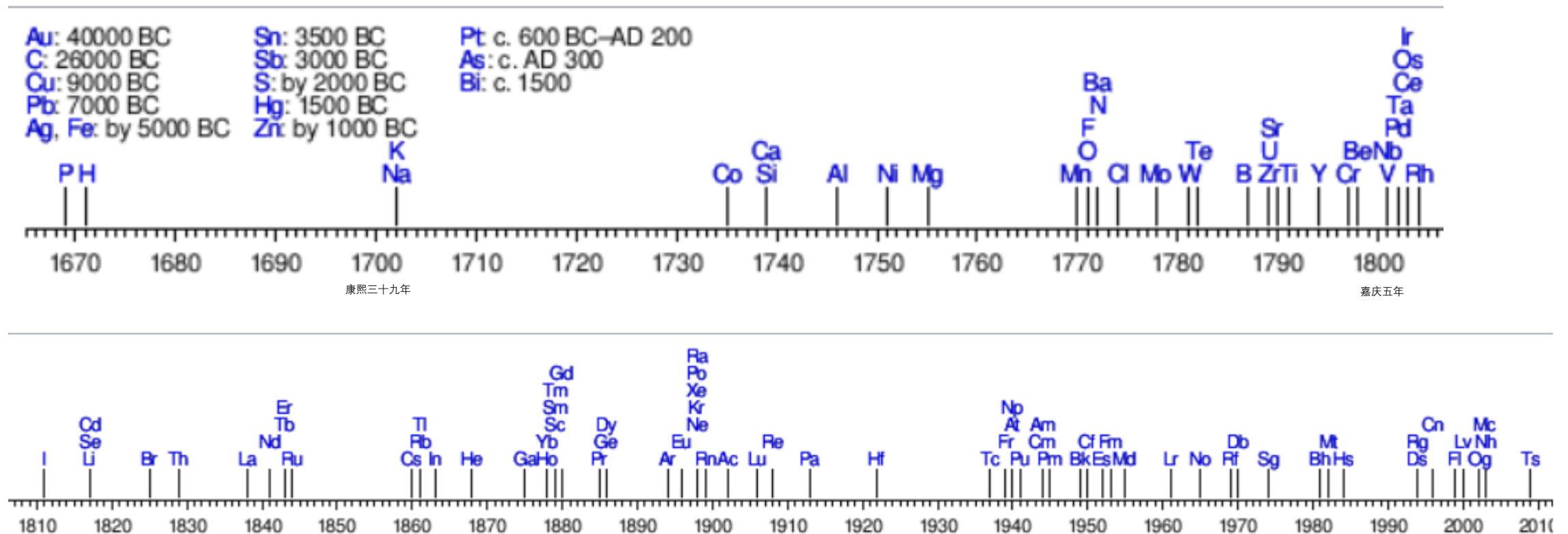
Fundamental Concepts in Chemistry

What is Chemistry?

- “Chemistry is the scientific study of the properties and behavior of matter. It is a physical science within the natural sciences that studies the chemical elements that make up matter and compounds made of atoms, molecules and ions: their composition, structure, properties, behavior and the changes they undergo during reactions with other substances. Chemistry also addresses the nature of chemical bonds in chemical compounds.” – Wikipedia
- “化学是在原子、分子水平上研究物质的组成、结构、性质、转化及其应用的基础自然科学。它源自生活和生产实践，并随着人类社会的进步而不断发展。不同于研究尺度更小的粒子物理学与核物理学，化学研究的原子、分子、离子（团）的物质结构和化学键、分子间作用力等相互作用，其所在的尺度是微观世界中最接近宏观的，因而它们的自然规律也与人类生存的宏观世界中物质和材料的物理、化学性质最为息息相关。作为沟通微观与宏观物质世界的重要桥梁，化学则是人类认识和改造物质世界的主要方法和手段之一。” – 百度百科

Discovery of Elements

- 14 elements have been discovered by 1600: Au, C, Cu, Pb, Ag, Fe, Sn, Sb, S, Hg, Zn, Pt, As, Bi



Pure Substances and Mixture

- Pure substance is any matter that has a fixed chemical composition and characteristic properties.
 - Element / elemental substance []: a pure substance which is composed of _____
 - Compound []: a pure substance which is composed of _____
- Mixture is combination of two/more substances in which the substances retain their distinct identities.
 - Homogeneous mixture
 - Heterogeneous mixture

Mixtures

- Mixtures are either homogenous or heterogeneous.

Homogeneous mixture	Heterogeneous mixture
All the components are in same state with no visible boundaries	Components may not be in same states. They may have visible boundaries
They are uniform in their composition	They may not be uniform in their composition
Separated by physical methods such as distillation and crystallization	Separated by physical methods such as filtration and decantation.

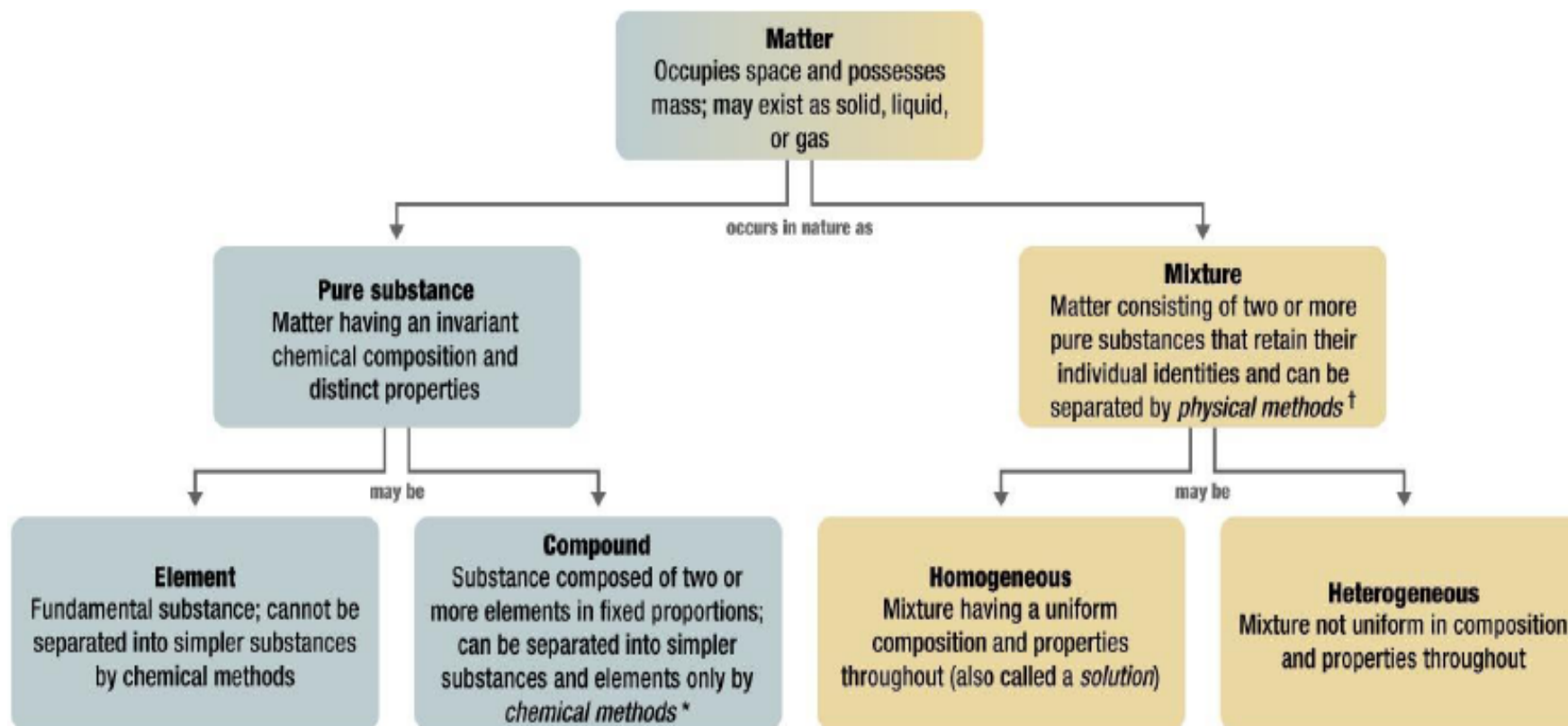
Colloids often look homogeneous to the eye but are technically heterogeneous under a microscope.

Common techniques to separate mixtures

What does each technique separate?

- Filtration
- Distillation
- Crystallization
- Solvent extraction
- Sublimation

Summary of Matter



* *Chemical methods of separation include electrolysis.*

† *Physical methods of separation include filtration, distillation, and crystallization.*

Properties of Matter

- Substances are identified by their properties.
- They are of two types - Physical and Chemical properties.
- Physical properties are characteristics that a scientist can measure without changing the composition of the sample.
 - Without chemical reaction
 - Eg. Mass, Color, Melting point, Boiling point
- Chemical properties describe the characteristic ability of a substance to react, hence forming a new substance.
 - With chemical reaction
 - Eg. CO_2 is bubbled into lime water

Intensive and Extensive properties

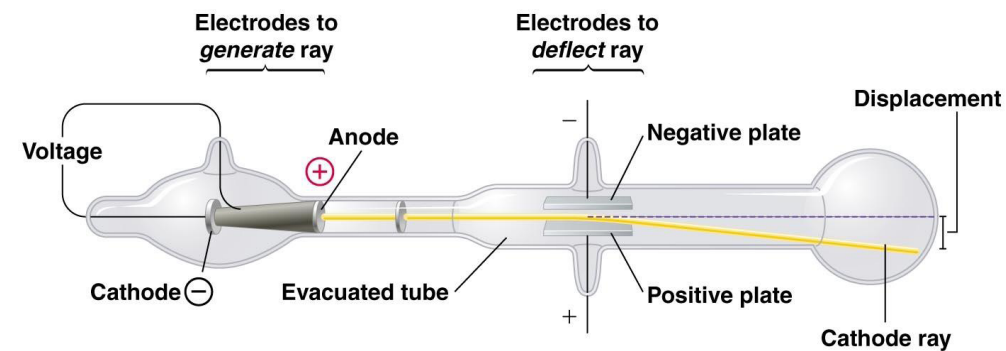
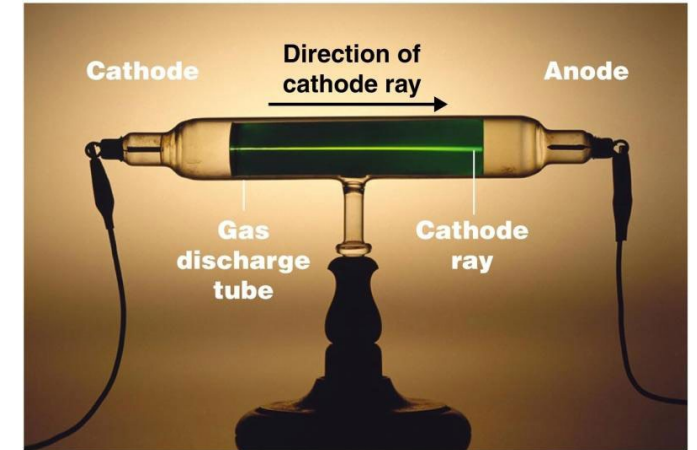
- Physical properties can be Extensive or Intensive.
 - Extensive Properties
 - vary with amount of the substance
 - mass, weight, volume, ...
 - convey information about the amount of substance in the sample
 - Intensive Properties
 - do not vary with amount of substance
 - color, melting point, boiling point, ...
 - convey the substance's identity
- Extensive or intensive property?
- Density
 - Number of particles
 - Temperature
 - Amount of substance (number of mole)

Part 2

Atomic Structure – Discovery and History

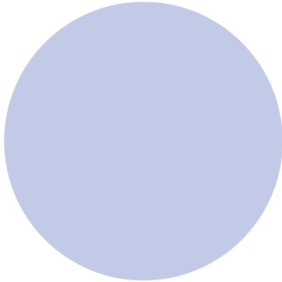
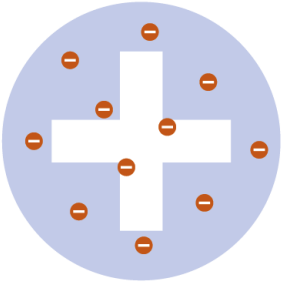
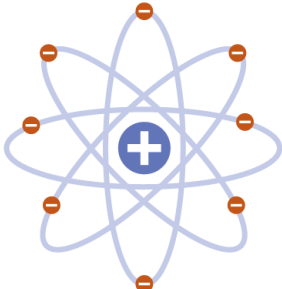
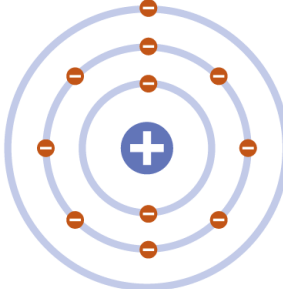
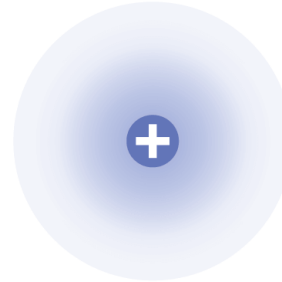
Discovery of atomic structure

- Which particle was discovered earliest / latest?
- Proton
- Neutron
- Electron

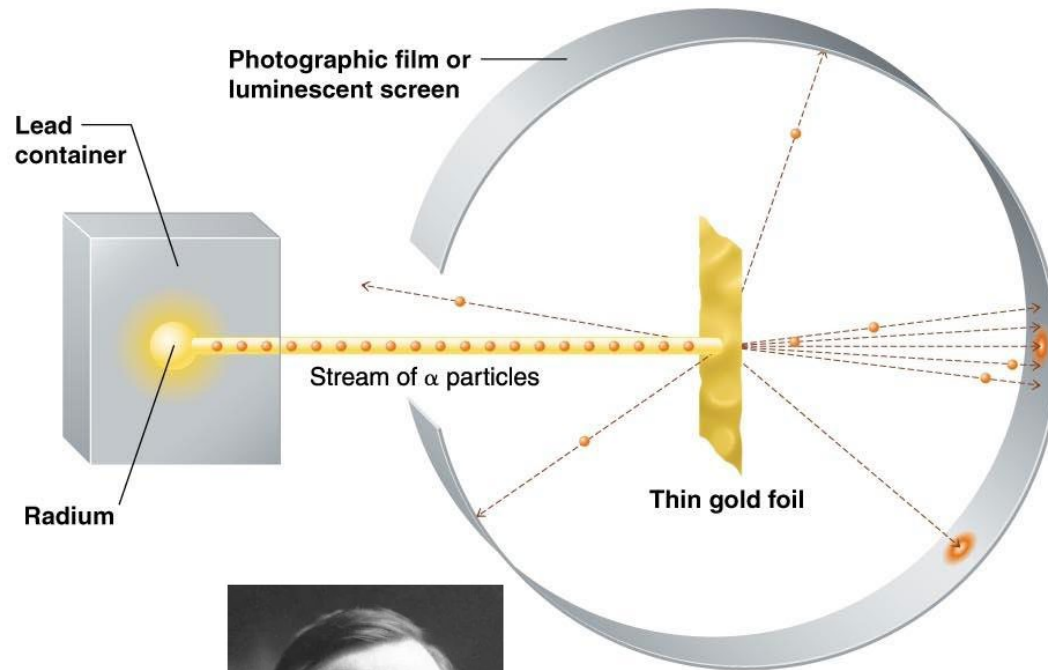


A History of the Atom: Theories and Models

How have our ideas about atoms changed over the years? This graphic looks at atomic models and how they developed.

Solid sphere model	Plum pudding model	Nuclear model	Planetary model	Quantum model
				
John Dalton	J.J. Thomson	Ernest Rutherford	Niels Bohr	Erwin Schrödinger
 1803	 1904	 1911	 1913	 1926
Dalton drew upon the Ancient Greek idea of atoms (the word 'atom' comes from the Greek 'atomos' meaning indivisible). His theory stated that atoms are indivisible, those of a given element are identical, and compounds are combinations of different types of atoms.	Thomson discovered electrons (which he called 'corpuscles') in atoms in 1897, for which he won a Nobel Prize. He subsequently produced the 'plum pudding' model of the atom. It shows the atom as composed of electrons scattered throughout a spherical cloud of positive charge.	Rutherford fired positively charged alpha particles at a thin sheet of gold foil. Most passed through with little deflection, but some deflected at large angles. This was only possible if the atom was mostly empty space, with the positive charge concentrated in the centre: the nucleus.	Bohr modified Rutherford's model of the atom by stating that electrons moved around the nucleus in orbits of fixed sizes and energies. Electron energy in this model was quantised; electrons could not occupy values of energy between the fixed energy levels.	Schrödinger stated that electrons do not move in set paths around the nucleus, but in waves. It is impossible to know the exact location of the electrons; instead, we have 'clouds of probability' called orbitals, in which we are more likely to find an electron.
 Recognised that atoms of a particular element differ from other elements.	 Recognised electrons as components of atoms.	 Realised that positive charge was localised in the nucleus of an atom.	 Proposed stable electron orbits; explained the emission spectra of some elements.	 Shows electrons don't move around the nucleus in orbits, but in clouds where their position is uncertain.
 Atoms aren't indivisible—they're composed from subatomic particles.	 No nucleus, and didn't explain later experimental observations.	 Did not explain why electrons remain in orbit around the nucleus.	 Moving electrons should emit energy and collapse into the nucleus; model did not work well for heavier atoms.	 Still widely accepted as the most accurate model of the atom.

Rutherford's Gold Foil Experiment



(a) Rutherford's experiment

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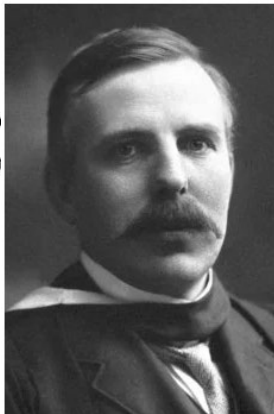
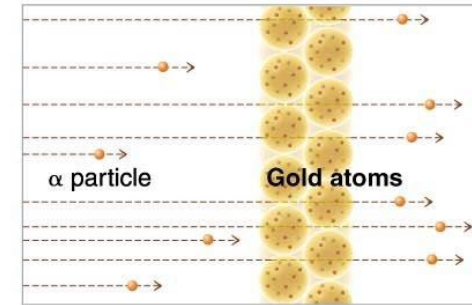


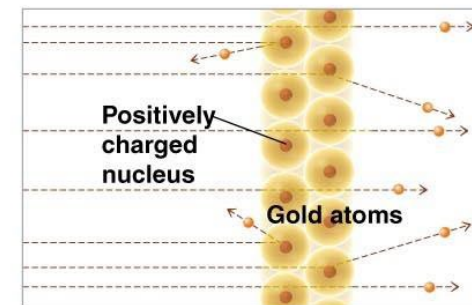
Photo from the Nobel Foundation archive.
Ernest Rutherford

Prize share: 1/1

The Nobel Prize in Chemistry 1908 was awarded to Ernest Rutherford "for his investigations into the disintegration of the elements, and the chemistry of radioactive substances"



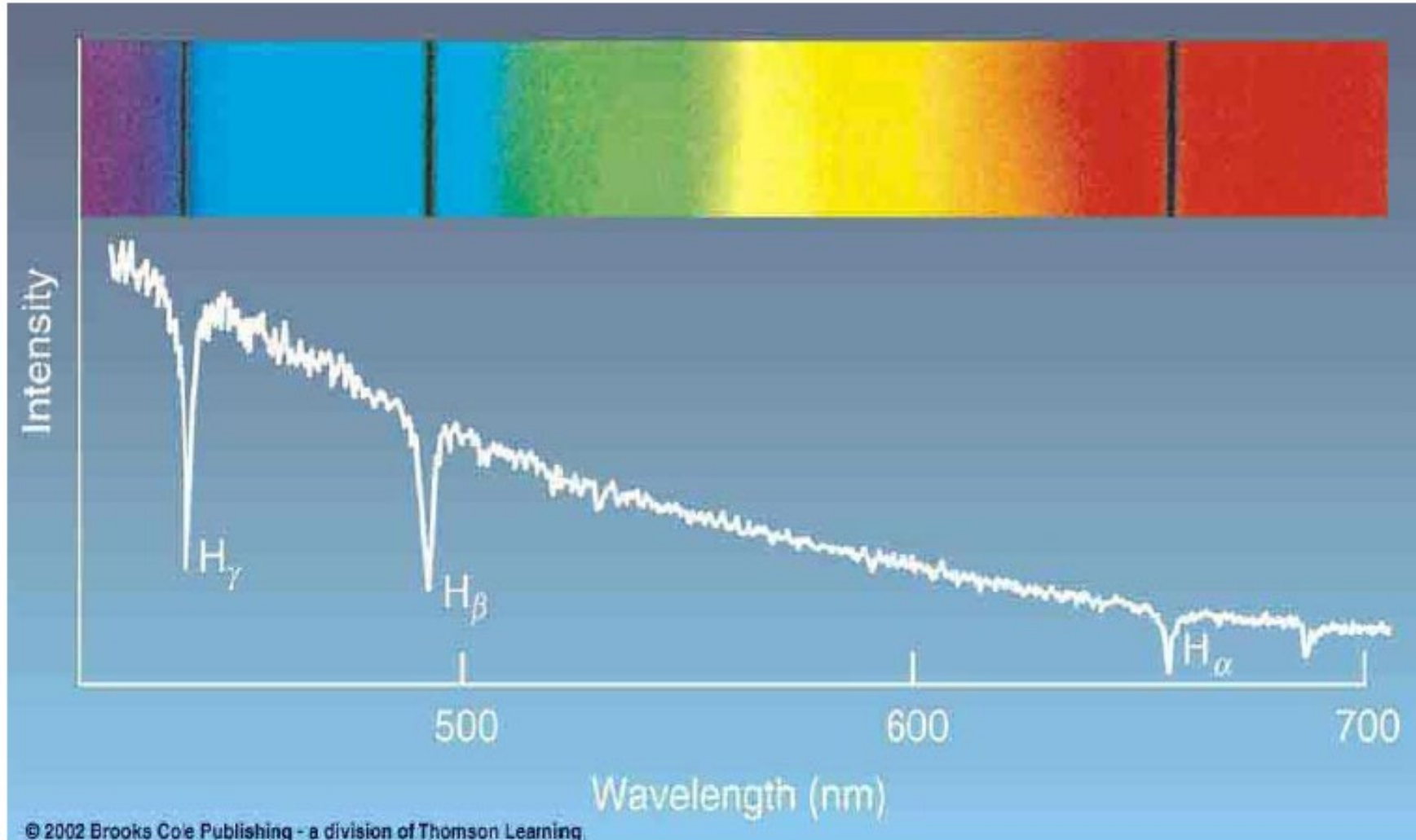
(b) What Rutherford expected if Thomson's model were correct



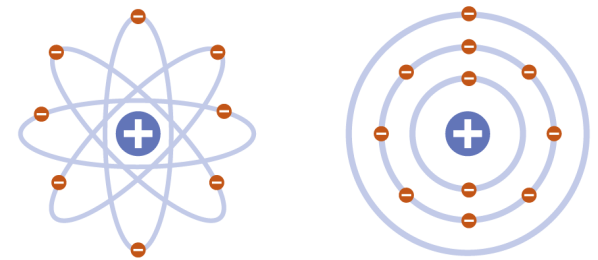
(c) What Rutherford actually observed

- ▶ Experiment: Rutherford aimed a stream of alpha particles at a very thin gold foil target and examined the scattering
- ▶ Observation:
 - (i) Majority passed through the foil in a straight line.
 - (ii) Some alpha particles were deflected at small angles
 - (iii) A very tiny fraction were reflected at large angles or even directly back at the source
- ▶ Conclusion: Both mass and the positive charge are concentrated in a tiny fraction of the volume of an atom called the nucleus.

Hydrogen atom is the simplest for modelling



- How to explain the absorption / emission spectrum of hydrogen?



Bohr's model of Hydrogen Atom

- Bohr proposed that electrons orbited around the nucleus in circles.
- Bohr fixed the orbital radii to certain discrete values.
- With this proposal, he could derive the energies of the orbits.



Photo from the Nobel Foundation archive.
Niels Henrik David Bohr
Prize share: 1/1

$$E_n = -\frac{\mu Z^2 e^4}{8 \epsilon_0^2 h^2} \frac{1}{n^2} \quad (2.1)$$

The Nobel Prize in Physics 1922 was awarded to Niels Henrik David Bohr "for his services in the investigation of the structure of atoms and of the radiation emanating from them"

E is the energy of the hydrogenic atom or ion (any atom or ion with one electron)

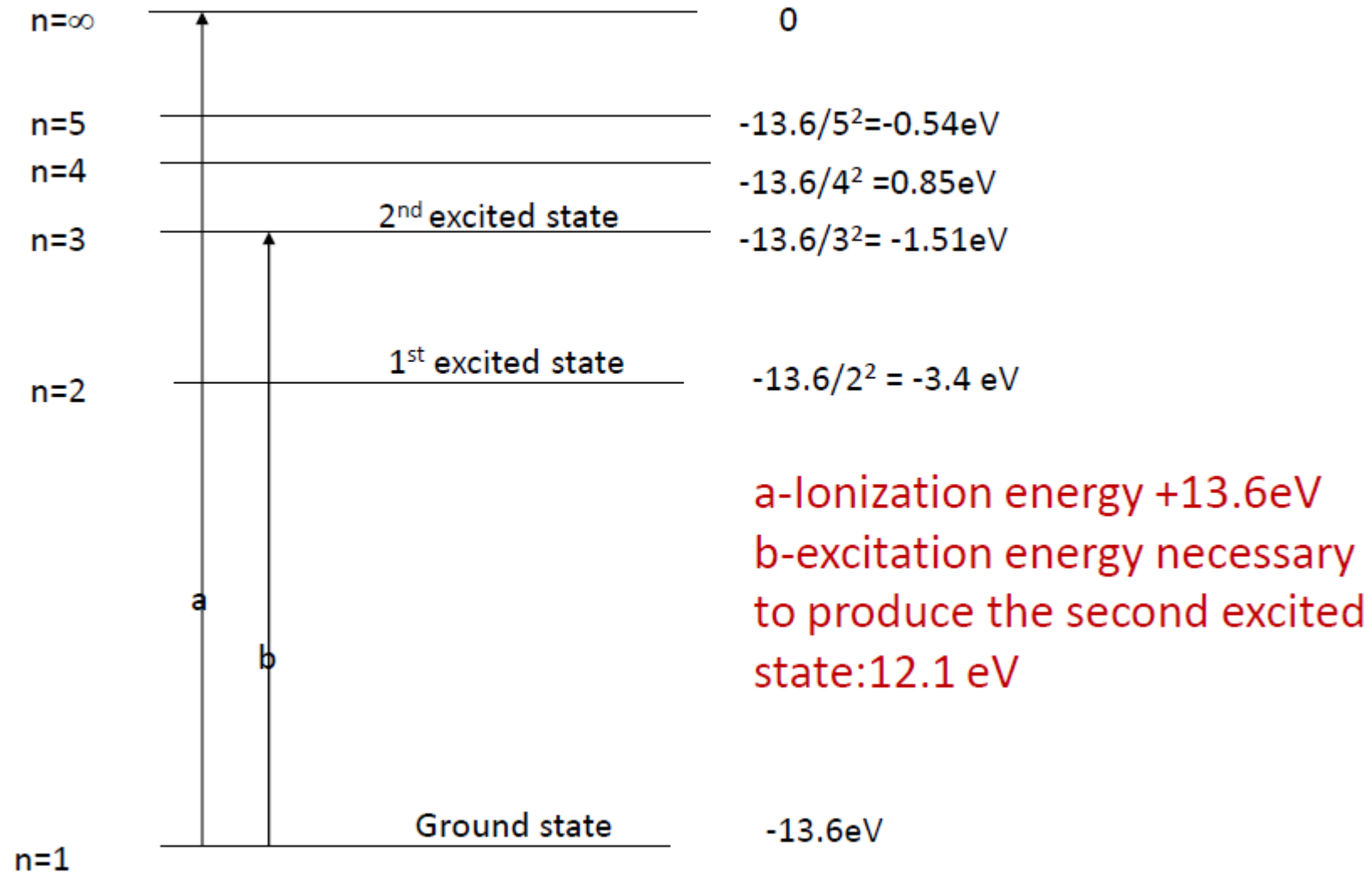
" Z " is the atomic number and " e " is the magnitude of the charge on the electron

μ is the reduced mass, approximately equal to the mass of the electron

ϵ_0 is the permittivity of free space

h is Planck constant

Energy level diagram for hydrogen atom



Issues with the Bohr's Atomic Model

- This model can ONLY explain the spectra of particles with ONE electron (H, He^+ , Li^{2+} , H_2^+ ...). Cannot explain particles with two and more electrons.
- According to classical electromagnetic theory, moving charged particle (electron) will emit electromagnetic radiation $\rightarrow$ radiation brings away energy $\rightarrow$ potential energy of electron decreases $\rightarrow$ electron eventually fall towards nucleus $\rightarrow$ atoms do not exist
- The experiments weren't wrong, therefore the theory needed to be modified.
- The only conclusion was the “planetary model” need improvement: electrons do not move in circular (nor elliptical) orbits around the nucleus.
- A better theory was needed.

The Schrödinger Wave Equation

- The “movement” of electrons surrounding the nucleus can be described by the wave function ψ
- ψ describes the probability of the electron to “appear” at a given position
- ψ can be visualized as “electron cloud”



Photo from the Nobel Foundation archive.



Photo from the Nobel Foundation archive.

- Erwin Schrödinger: Nobel Prize in Physics 1933
- Werner Karl Heisenberg: Nobel Prize in Physics 1932

$$\hat{H}\psi_n = E_n\psi_n$$

$\hat{H}$ is Hamiltonian operator.

- ψ is the wave function.
- E is the total energy of the system under consideration – it is simply a number.

According to Heisenberg's Uncertainty Principle:
it is impossible to accurately measure the position and the movement of a particle

$$\Delta x \Delta p \geq \frac{h}{4\pi}$$

单电子、一维、非相对论薛定谔方程

$$i\hbar \frac{\partial}{\partial t} \Psi(x, t) = \left[-\frac{\hbar^2}{2m} \frac{\partial^2}{\partial x^2} + V(x, t) \right] \Psi(x, t)$$

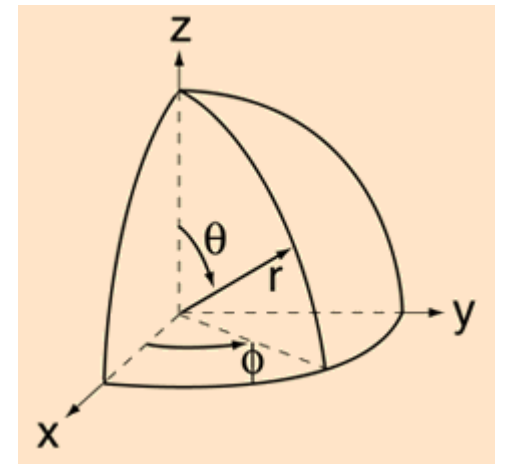
Solutions to the Schrödinger Equation

- Schrödinger found that it is easier to solve when we change from *x-y-z coordinates (直角坐标系)* into *spherical polar coordinates (球坐标系)*.
- ψ could be factored into a radial part $R(r)$ and an angular part $Y(\theta, \phi)$.

$$\psi_{n,l,m_l}(r, \theta, \phi) = R_{n,l}(r) Y_{l,m_l}(\theta, \phi)$$

- Bohr found that only n was needed, but upon solving the Schrödinger equation it was found that...three integers, denoted as n, l, m_l are needed to solve the Schrödinger equation.

Good news is that mathematicians have solved the equation and chemists just need to make use of the solution.



Nobel Prizes Related to Atomic Structure

Year	Laureate(s)	Field	Contribution
1908	Ernest Rutherford	Physics	Investigations into the structure of the atom; discovery of the atomic nucleus
1911	Marie Curie	Physics	Studies on radioactivity related to atomic decay
1913	Niels Bohr	Physics	Model of atomic structure and quantum theory of the atom
1915	William Henry Bragg & William Lawrence Bragg	Physics	X-ray crystallography revealing atomic arrangements
1922	Niels Bohr	Physics	Quantum theory of the atom
1923	Robert Millikan	Physics	Measurement of the electron's charge
1925	James Franck & Gustav Hertz	Physics	Experimental proof of quantized atomic energy levels
1926	Jean Perrin	Physics	Experimental proof of atomic reality
1932	Werner Heisenberg	Physics	Quantum mechanics applied to atomic structure (matrix mechanics)
1932	James Chadwick	Physics	Discovery of the neutron

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1933	Erwin Schrödinger & Paul Dirac	Physics	Wave mechanics and quantum mechanics equations describing atomic behavior
1935	Frédéric Joliot & Irène Joliot-Curie	Chemistry	Artificial radioactivity related to atomic nuclei
1944	Isidor Isaac Rabi	Physics	Resonance method for recording magnetic properties of atomic nuclei
1954	Max Born & Walther Bothe	Physics	Fundamental work in quantum mechanics and nuclear reactions
1961	Robert Hofstadter	Physics	Studies of nucleon structure
1962	Lev Landau	Physics	Theoretical work on condensed matter, connected to atomic interactions
1965	Richard Feynman, Julian Schwinger, Sin-Itiro Tomonaga	Physics	Quantum electrodynamics explaining atomic interactions
1986	Dudley Herschbach, Yuan Lee, John Polanyi	Chemistry	Dynamics of chemical elementary processes at atomic/molecular level
1997	Steven Chu, Claude Cohen-Tannoudji, William D. Phillips	Physics	Laser cooling and trapping of atoms
2001	Wolfgang Ketterle, Eric Cornell, Carl Wieman	Physics	Bose-Einstein condensate (atomic behavior at ultra-cold temperatures)

Part 3

Atomic Structure – Modern model by Quantum Mechanics

Quantum Numbers n , l , and m_l

Quantum number	Name	Value	Physical meaning
n 主量子数	Principle quantum number	1, 2, 3, ...	The energy level of electrons. The shell that the electron is located.
l 角量子数	Azimuthal quantum number (orbital angular momentum quantum number)	0, 1, 2, ..., (n-1)	The distribution of electron cloud in space. The shape of the orbital.
m_l 磁量子数	Magnetic quantum number	0, ± 1 , ± 2 , ..., $\pm l$	Orientation (direction) of the orbital.

Allowed Quantum numbers

TABLE 6.2		Allowed Values of the Quantum Numbers n , ℓ , and m_ℓ	
When n is	ℓ can be	When ℓ is	m_ℓ can be
1	only 0	0	only 0
2	0 or 1	0 1	only 0 -1, 0, or +1
3	0, 1, or 2	0 1 2	only 0 -1, 0, or +1 -2, -1, 0, +1, or +2
4	0, 1, 2, or 3	0 1 2 3	only 0 -1, 0, or +1 -2, -1, 0, +1, or +2 -3, -2, -1, 0, +1, +2, or +3
.	.	.	.
.	.	.	.
.	.	.	.

Allowed Values of the Quantum Numbers

- Will denote here as (n, l, m_l)
- Some allowed values:
 - $(1,0,0)$, $(5,4,-1)$, $(2,1,0)$, $(2,0,0)$, $(4,3,3)$
- Some impossible values:
 - $(1,1,0)$, $(5,4,-5)$, $(2,1,-2)$, $(2,0,1)$, $(0,3,3)$, $(0,0,0)$
- By convention, lower-case letters have been used to designate the l *quantum number*
- What is the symbol of the orbitals?
 - $(1, 0, 0)$
 - $(5, 4, -1)$
 - $(4, 3, 3)$

l	Letter
0	s
1	p
2	d
3	f
4	g
5	h

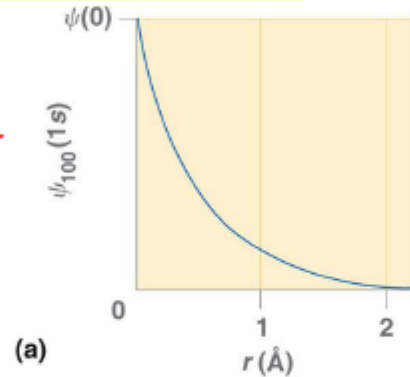
How does a 1s orbital look like?

The functions, ψ_{n,l,m_l} are called orbitals (rather than orbits).

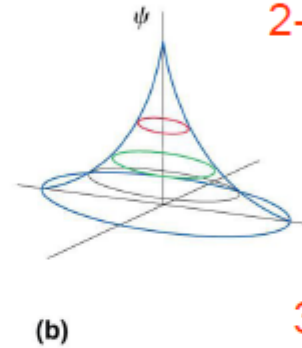
$$\psi_{1,0,0}(r, \theta, \phi) = \frac{1}{\sqrt{\pi}} a_0^{-3/2} e^{-r/a_0}$$

a_0 is the “Bohr radius” ≈ 52.9 pm

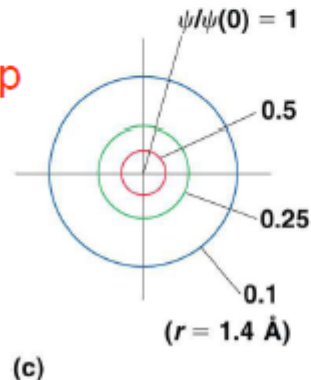
1-D ψ vs r



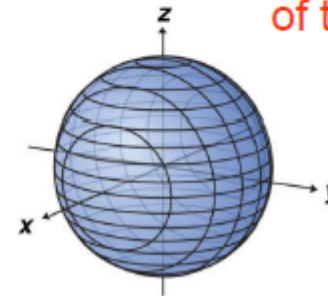
2-D ψ vs r



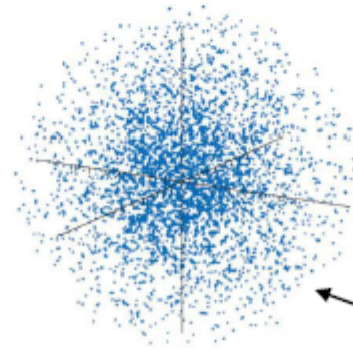
2-D ψ vs r
contour map



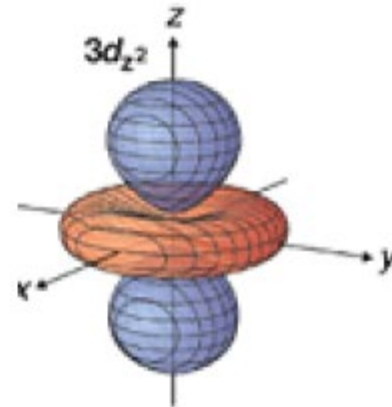
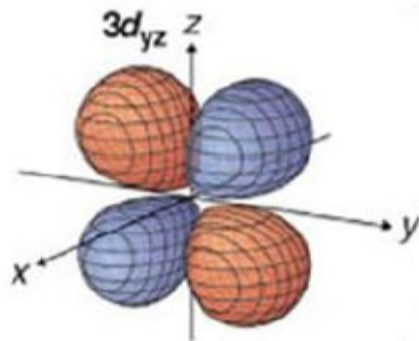
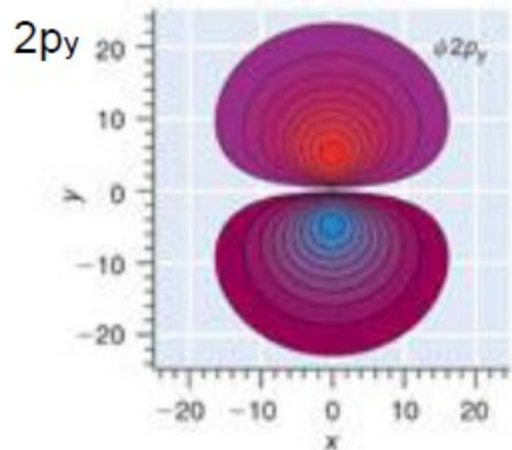
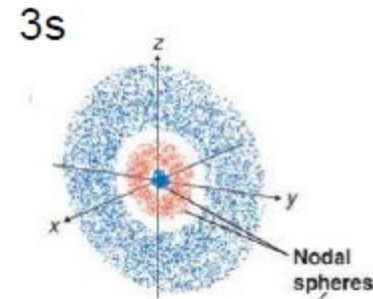
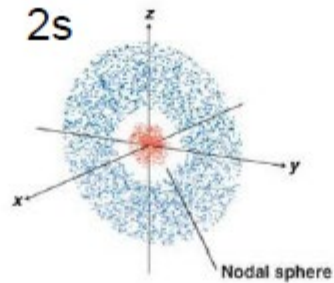
3-D Boundary Surface
Diagram enclosing most
of the cloud



3-D Cloud

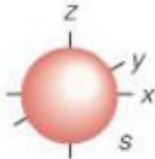
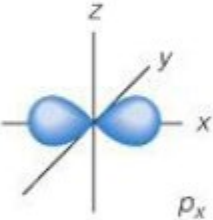
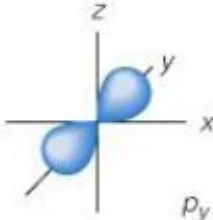
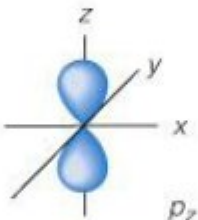
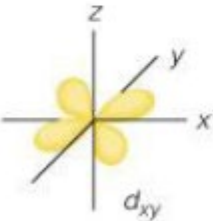
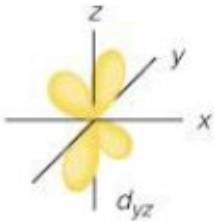
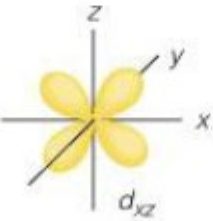
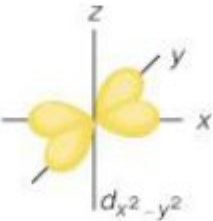
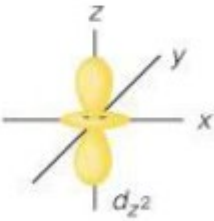


How does other orbitals look like?



- Node (planar or spherical)
 - “Restrict” the movement of electrons
 - Increase / decrease the energy of electron

Sub-levels and orbitals

Sublevel	Orbital Shapes	Orbitals per Sublevel	Lobes per Orbital
s		1	1
p	  	3	2
d	    	5	4*
f	Seven complex shapes	7	8*

Summary of Quantum numbers

principal (n) size
angular (l) shape
magnetic (m_l) orientation

} Required to describe an atomic orbital

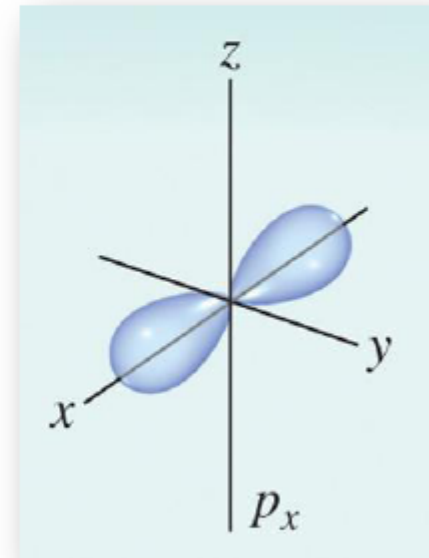
principal ($n = 2$)

$2p_x$

related to the
magnetic quantum
number (m_l)

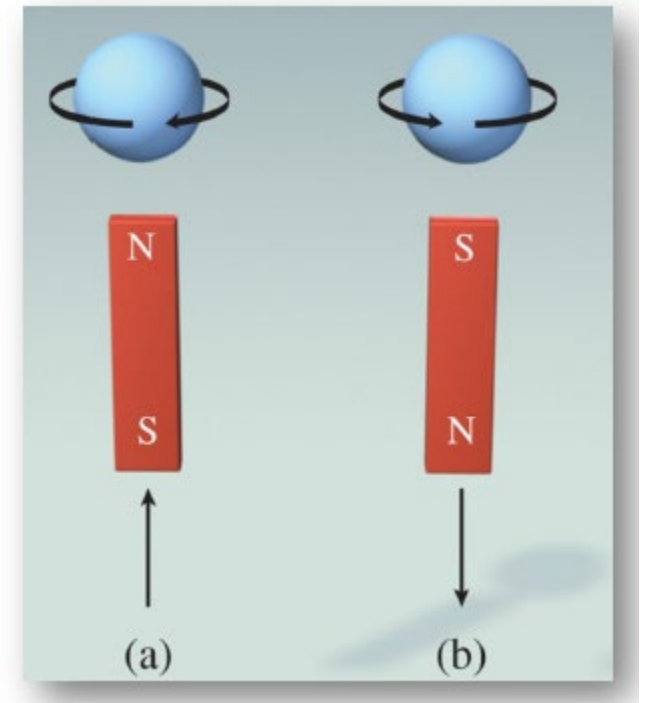
angular momentum ($l = 1$)

electron spin (m_s) direction of spin



Spin magnetic quantum number m_s

- Each orbital can contain zero, one or two electrons.
- Two electron in one orbital must have opposite spins.
- This is specified by the spin quantum number (m_s).
- There are two possible directions of spin.
- Allowed values of m_s are $+\frac{1}{2}$ and $-\frac{1}{2}$.



Pauli's Exclusion Principle

- No two electrons in a given atom can have exactly the same set of quantum numbers.
- For the He atom, the configuration $1s^2$ means that
 - one electron has the quantum numbers $n = 1, l = 0, m_l = 0, m_s = +\frac{1}{2}$,
 - the second electron: $n = 1, l = 0, m_l = 0, m_s = -\frac{1}{2}$.
- Li could not have the configuration $1s^3$, since the third electron would have to get the same set of quantum numbers as one of the other two.

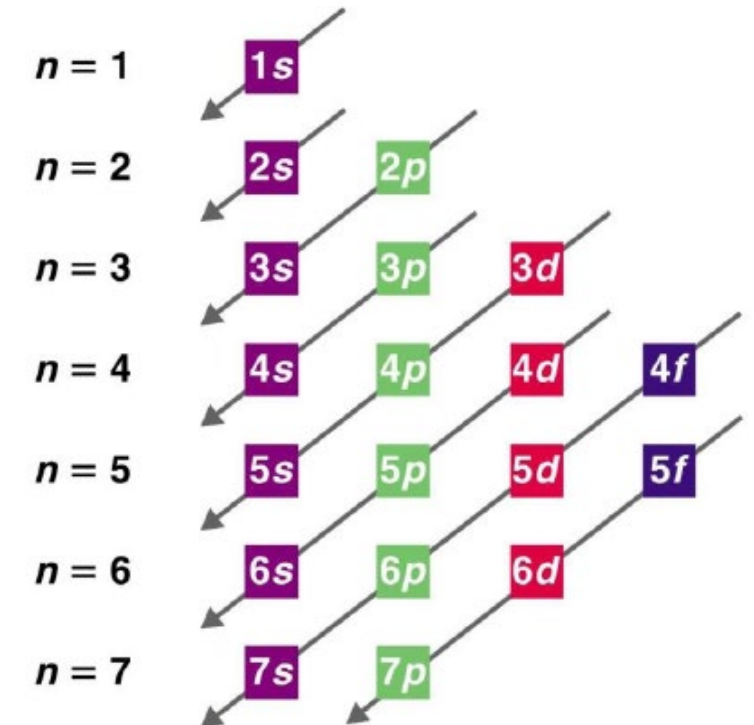
Aufbau, from German, meaning “building-up principle”.

构造原理

能量最低原则

The Aufbau, or Building-Up, Principle

- A rule for finding the lowest energy electronic configuration for atoms.
- It states that you must *add the electrons, one at a time, into the orbitals that would make the atom most stable, also obeying the Pauli exclusion principle at all times.*
- By “most stable”, we mean “lowest value to total electronic energy, E , of all electrons”.



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Stable Electronic Configurations

- The order of filling AO is best remembered by picturing the periodic table in your mind.
- The table to the right illustrates this

You should be able to give ground state (most stable state) electronic configurations for any element in the periodic table given the Z only.

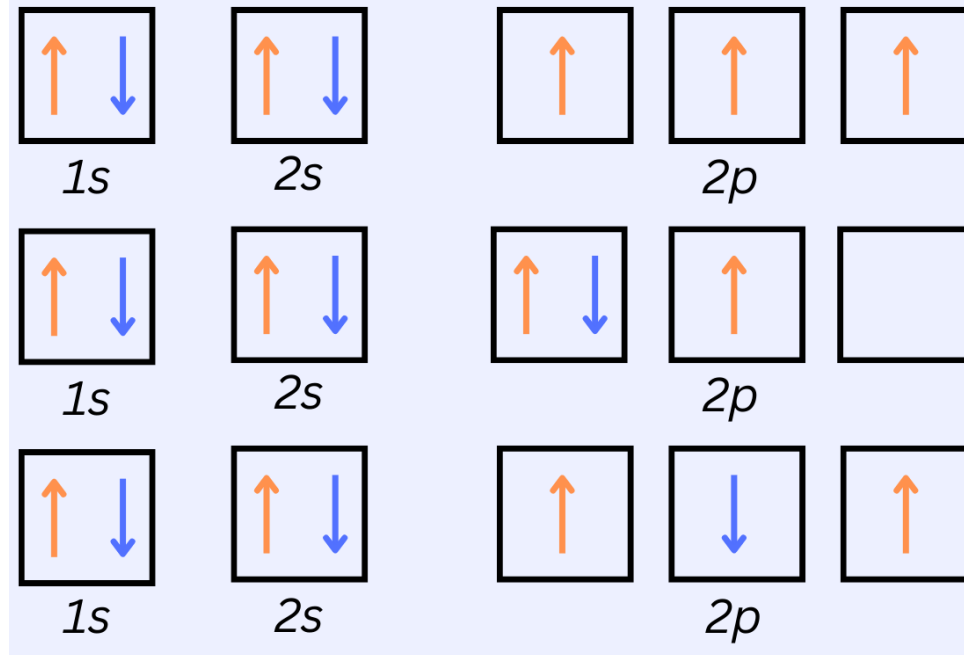
Orbitals Filled			
1s			
2s			2p
3s			3p
4s		3d	4p
5s		4d	5p
6s	4f	5d	6p
7s	5f	6d	7p

ground state (most stable state) electronic configurations for any element in the periodic table given the Z only.

IA		IIA		Transition metals																IIIA	IVA	VA	VIA	VIIA	0								
1	2																	5	6	7	8	9	10										
H	He																	B	C	N	O	F	Ne										
3	4																	13	14	15	16	17	18										
Li	Be																	Al	Si	P	S	Cl	Ar										
11	12																	21	22	23	24	25	26	27	28	29	30	31	32	33	34	35	36
Na	Mg																	Sc	Ti	V	Cr	Mn	Fe	Co	Ni	Cu	Zn	Ga	Ge	As	Se	Br	Kr
19	20																	39	40	41	42	43	44	45	46	47	48	49	50	51	52	53	54
K	Ca																	Y	Zr	Nb	Mo	Tc	Ru	Rh	Pd	Ag	Cd	In	Sn	Sb	Te	I	Xe
37	38	Lanthanides / Actinides																71	72	73	74	75	76	77	78	79	80	81	82	83	84	85	86
Rb	Sr																	Lu	Hf	Ta	W	Re	Os	Ir	Pt	Au	Hg	Tl	Pb	Bi	Po	At	Rn
55	56	57	58	59	60	61	62	63	64	65	66	67	68	69	70	71	72	73	74	75	76	77	78	79	80	81	82	83	84	85	86		
Cs	Ba	La	Ce	Pr	Nd	Pm	Sm	Eu	Gd	Tb	Dy	Ho	Er	Tm	Yb	Lu	Hf	Ta	W	Re	Os	Ir	Pt	Au	Hg	Tl	Pb	Bi	Po	At	Rn		
87	88	89	90	91	92	93	94	95	96	97	98	99	100	101	102	103	104	105	106	107	108	109	110	111	112	113	114	115	116	117	118		
Fr	Ra	Ac	Th	Pa	U	Np	Pu	Am	Cm	Bk	Cf	Es	Fm	Md	No	Lr	Rf	Db	Sg	Bh	Hs	Mt	Uun	Uuu	Uub	Uut	Uuq	Uur	Uus	Uuh	Uuo		

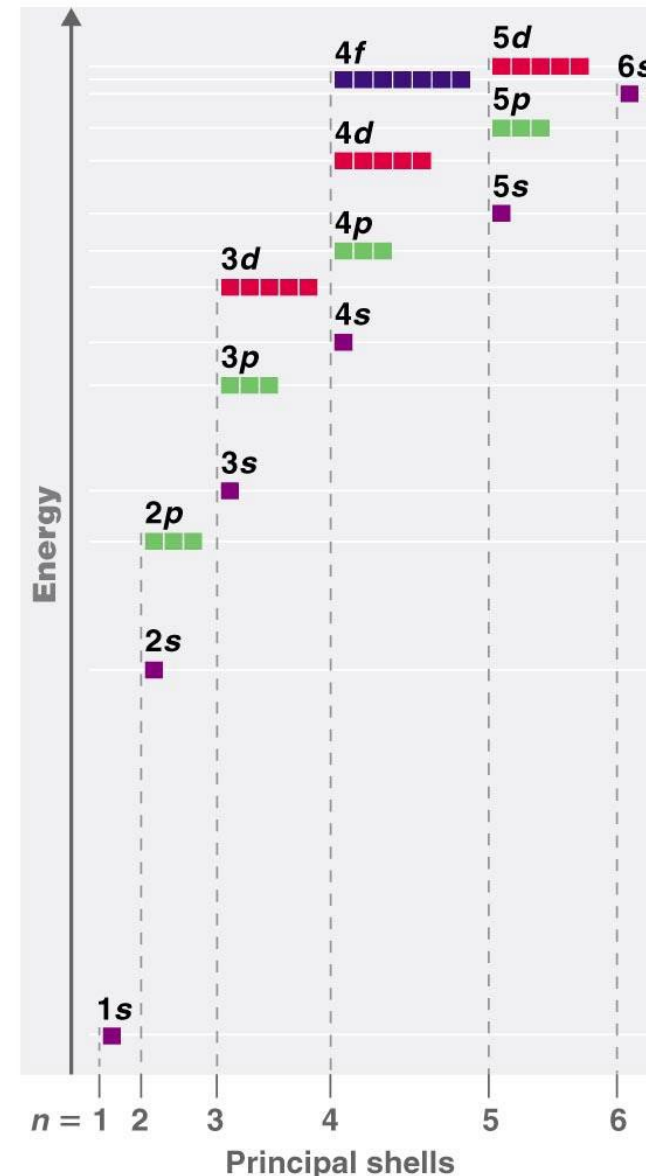
Hund's Rule of Maximum Multiplicity

- Part 1: Degenerate orbitals ($l > 0$) are *always filled with single electrons before any of them are doubly filled*.
- Part 2: Electrons prefer to fill the separate orbitals with their spins aligned (same value of m_s).



General Rules for writing electron configurations

1. Electrons will reside in the available orbitals of the lowest possible energy
2. Each orbital can accommodate a maximum of two electrons
3. Electrons will occupy degenerated orbitals singly, before pairing up
4. Orbitals will fill in the order indicated in the figure



Isotope

- Distinct nuclear species of the same chemical element, with the same atomic number (proton number) but different nucleon numbers (mass numbers)
- Symbols for isotope:
 - Isotopes of hydrogen
 - Isotopes of carbon
 - Isotopes of uranium
 - Electron
 - Proton
 - Neutron



Periodic Table of Elements with Relative Atomic Masses

1																	18	
1 H 1.008	2											13	14	15	16	17	2 He 4.003	
3 Li 6.941	4 Be 9.012											5 B 10.81	6 C 12.01	7 N 14.01	8 O 16.00	9 F 19.00	10 Ne 20.18	
11 Na 22.99	12 Mg 24.31	3	4	5	6	7	8	9	10	11	12	13 Al 26.98	14 Si 28.09	15 P 30.97	16 S 32.07	17 Cl 35.45	18 Ar 39.95	
19 K 39.10	20 Ca 40.08	21 Sc 44.96	22 Ti 47.87	23 V 50.94	24 Cr 52.00	25 Mn 54.94	26 Fe 55.85	27 Co 58.93	28 Ni 58.69	29 Cu 63.55	30 Zn 65.38	31 Ga 69.72	32 Ge 72.64	33 As 74.92	34 Se 78.96	35 Br 79.90	36 Kr 83.80	
37 Rb 85.47	38 Sr 87.62	39 Y 88.91	40 Zr 91.22	41 Nb 92.91	42 Mo 95.96	43 Tc [98]	44 Ru 101.07	45 Rh 102.91	46 Pd 106.42	47 Ag 107.87	48 Cd 112.41	49 In 114.82	50 Sn 118.71	51 Sb 121.76	52 Te 127.60	53 I 126.90	54 Xe 131.29	
55 Cs 132.91	56 Ba 137.33	57 La 138.91	72 Hf 178.49	73 Ta 180.95	74 W 183.84	75 Re 186.21	76 Os 190.23	77 Ir 192.22	78 Pt 195.08	79 Au 196.97	80 Hg 200.59	81 Tl 204.38	82 Pb 207.2	83 Bi 208.98	84 Po (209)	85 At (210)	86 Rn (222)	
87 Fr (223)	88 Ra 226.0	89 Ac (227)	104 Rf (261)	105 Db (262)	106 Sg (269)	107 Bh (270)	108 Hs (270)	109 Mt (278)	110 Ds (281)	111 Rg (282)	112 Cn (285)	113 Nh (286)	114 Fl (289)	115 Mc (290)	116 Lv (293)	117 Ts (294)	118 Og (294)	
		58 Ce 140.12	59 Pr 140.91	60 Nd 144.24	61 Pm (145)	62 Sm 150.36	63 Eu 151.96	64 Gd 157.25	65 Tb 158.93	66 Dy 162.50	67 Ho 164.93	68 Er 167.26	69 Tm 168.93	70 Yb 173.05	71 Lu 174.97			
		90 Th 232.04	91 Pa 231.04	92 U 238.03	93 Np 237.05	94 Pu (244)	95 Am (243)	96 Cm (247)	97 Bk (247)	98 Cf (251)	99 Es (254)	100 Fm (257)	101 Md (256)	102 No (254)	103 Lr (257)			
40																		